데이터 로딩 + 확인

```
# 데이터 로딩

import pandas as pd

df = pd.read_csv('Chemical_Numeric_Data_Quality.csv', encoding='cp949')
```

```
# 가독성 향상을 위해 소수점 출력 포맷을 소수점 넷째자리까지 나오도록 설정
pd.set_option('display.float_format', lambda x: '%.4f' % x)
```

```
df.sample(3)
```

숨겨진 출력 표시

```
# 데이터 확인

df.info()
df.describe()
```

숨겨진 출력 표시

```
df.columns = ['Lot', 'Temp', 'Viscosity', 'Failure', 'FailureCode']
```

```
df.head()
```

숨겨진 출력 표시

```
df.FailureCode = df.FailureCode.fillna('None').astype(str)
```

```
df.FailureCode.value_counts()
```

숨겨진 출력 표시

```
df.FailureCode = df.FailureCode.apply(lambda x: x if x=='None' else str(int(float(x))))
```

```
df.sample(5)
```

숨겨진 출력 표시

```
df.info()
```

숨겨진 출력 표시

```
df.describe()
```

숨겨진 출력 표시

STEP ① 완전성 지표

```
# 결측치 확인
df.isnull().sum()
```

숨겨진 출력 표시

```
# 결측치 처리
df.dropna(how='any', axis=0, inplace=True)

df
```

숨겨진 출력 표시

STEP ② 유효성 지표

```
# 2. 유효성지표
# 이상치

outlier = df[(df.Temp < 20) | (df.Temp > 30) | (df.Viscosity > 3)]
```

```
outlier
```

	Lot	Temp	Viscosity	Failure	FailureCode
0	2110262249	24.4400	101.8000	1	2
1	2109975633	23.9500	82.6300	1	None
4	2109169908	23.3300	71.9200	1	None
6	2109032490	54.1100	2.1000	정상	None
7	2109332619	24.3900	71.1400	1	None
...	...	...	...	...	...
938	2111185019	26.3600	3.0100	정상	None
947	2111211907	23.6100	3.5500	정상	None
951	2111214194	27.1000	41.0500	불량	None
961	2111224103	24.0500	42.4500	1	None
980	2111256812	44.2200	51.6800	불량	None

296 rows × 5 columns

```
df.drop(outlier.index, axis=0, inplace=True)
```

STEP ③ 일관성 지표

```
df.Failure.value_counts()
```

	count
Failure	
0	412
정상	45
1	33
불량	5

dtype: int64

```
# df['Temp']
# df.loc[5]

df.Failure = df.Failure.map({'정상': 0, '0': 0,
                             '불량': 1, '1': 1})

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 495 entries, 2 to 986
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Lot          495 non-null    int64  
 1   Temp         495 non-null    float64
 2   Viscosity    495 non-null    float64
 3   Failure      495 non-null    int64  
 4   FailureCode  495 non-null    object 
dtypes: float64(2), int64(2), object(1)
memory usage: 23.2+ KB
```

```
df.Failure.value_counts()
```

	count
Failure	
0	457
1	38

dtype: int64

STEP ④ 유일성 지표

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 495 entries, 2 to 986
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Lot          495 non-null    int64  
 1   Temp         495 non-null    float64
 2   Viscosity    495 non-null    float64
 3   Failure      495 non-null    int64  
 4   FailureCode  495 non-null    object 
dtypes: float64(2), int64(2), object(1)
memory usage: 23.2+ KB
```

```
# 중복값이 있는지 확인
df.Lot.duplicated().sum()
```

```
np.int64(40)
```

```
# 중복값 삭제
df = df.drop_duplicates(subset=['Lot'], keep='first')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 455 entries, 2 to 975
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Lot          455 non-null    int64  
 1   Temp         455 non-null    float64
 2   Viscosity    455 non-null    float64
 3   Failure      455 non-null    int64  
 4   FailureCode  455 non-null    object 
dtypes: float64(2), int64(2), object(1)
memory usage: 21.3+ KB
```

STEP ⑤ 정확성 지표

```
df.sample(5)
```

	Lot	Temp	Viscosity	Failure	FailureCode
692	2110649000	26.7900	1.5400	0	None
423	2109458048	24.3500	2.4900	0	None
863	2110581504	27.0800	0.8700	0	None
320	2110580603	25.7000	2.8800	0	None
724	2111228330	25.8000	1.4100	0	None

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 455 entries, 2 to 975
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Lot          455 non-null    int64  
 1   Temp         455 non-null    float64
 2   Viscosity    455 non-null    float64
 3   Failure      455 non-null    int64  
 4   FailureCode  455 non-null    object 
dtypes: float64(2), int64(2), object(1)
memory usage: 21.3+ KB
```

```
# CASE1
incorrect_condi1_df = df[(df.Failure == 0) & (df.FailureCode != 'None')]
df.loc[incorrect_condi1_df.index, 'FailureCode'] = 'None'

# CASE2
incorrect_condi2_df = df[(df.Failure == 1) & (df.FailureCode == 'None')]
df.drop(incorrect_condi2_df.index, axis=0, inplace = True)
```

숨겨진 출력 표시

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 449 entries, 2 to 975
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Lot          449 non-null    int64  
 1   Temp         449 non-null    float64
 2   Viscosity    449 non-null    float64
 3   Failure      449 non-null    int64  
 4   FailureCode  449 non-null    object 
dtypes: float64(2), int64(2), object(1)
memory usage: 21.0+ KB
```